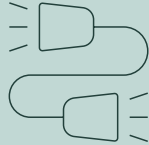


The Language of Networks

An Introduction to IEEE 802.3 and 802.11 Standards

IEEE 802.3
Standard for wired
Ethernet
communication.



IEEE 802.11
Standard for wireless
local networking.

ETHERNET

- Ethernet (pronounced “eether net”) is a **most widely used LAN technology** , which is defined under IEEE standards **802.3**.
- Ethernet network is used to create local area network and connect multiple computers or other devices such as printers, scanners, and so on.
- In a **wired network**, this is done with the help of **fiber optic cables**, while in a **wireless network**, it is done through **wireless network technology**.
- An Ethernet network **uses various topologies such as star, bus, ring, and more**.
- Ethernet connecting computers together with cable so the computers can share information.
- Ethernet operates in the data link layer and the physical layer.

ETHERNET

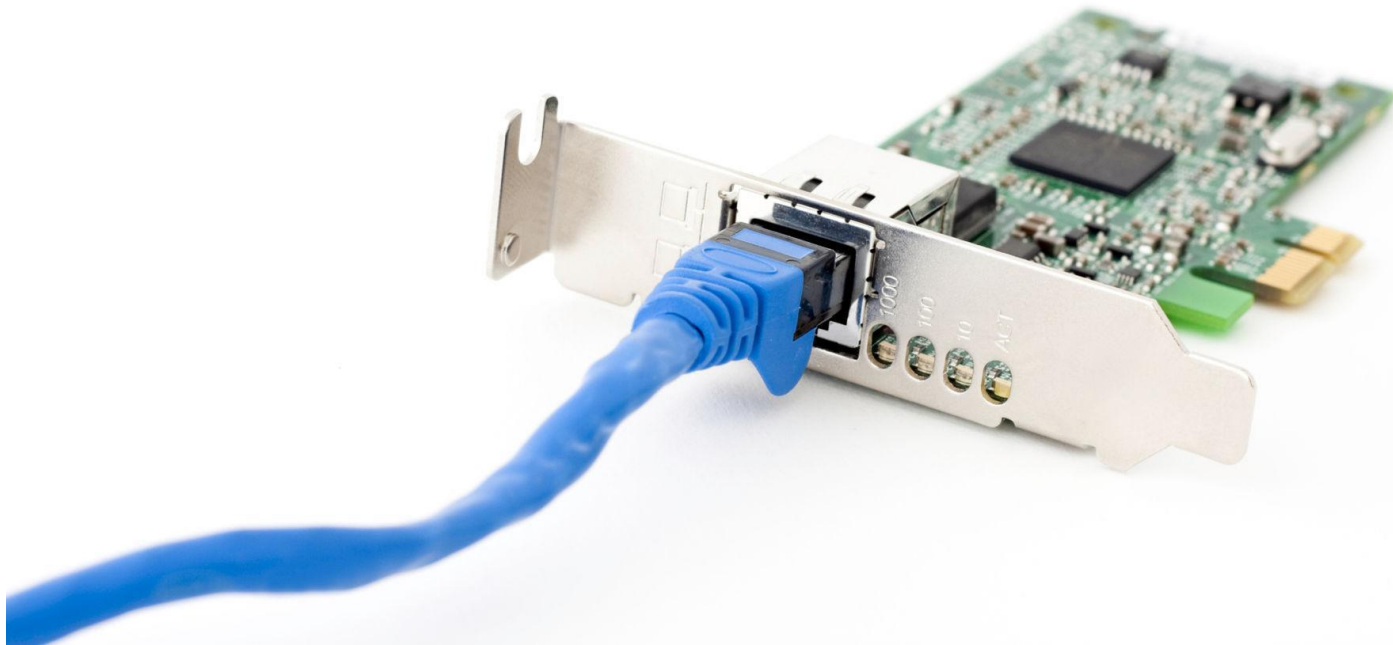
Wired Ethernet:

- Wired Ethernet network, devices are connected with the help of a **fiber optic cable** which connects the devices **within a distance of 10km**.
- For this, we have to **install a computer network interface card (NIC) in each computer**.
- A **unique address is given to each computer** that is connected.
- Ethernet is a **shared medium network technology**, where all the workstations are connected to the same cable and must connect with one another to send signals over it.

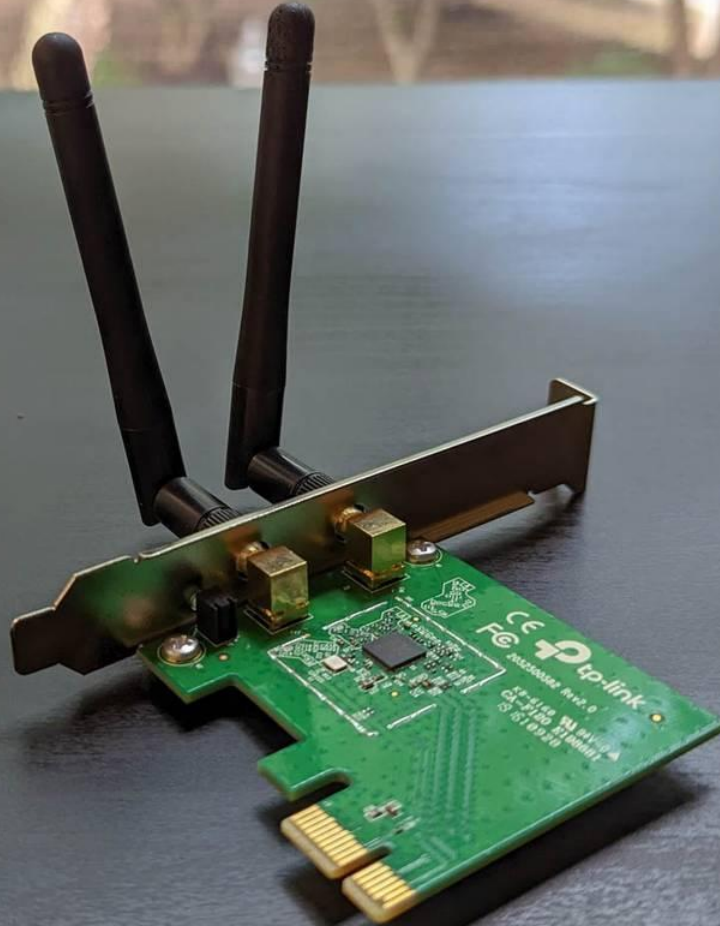
Wireless Ethernet

- In this, **wireless NICs are used** for connecting the computer instead of a cable.
- These wireless NICs make **use of radio waves** for communicating between the systems and furthers these NICs are connected with a wireless switch or hub.

Wired Network Interface Card

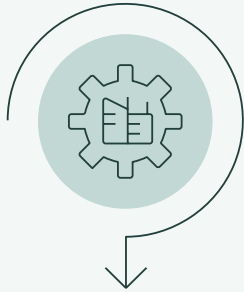


Wireless Network Interface Card



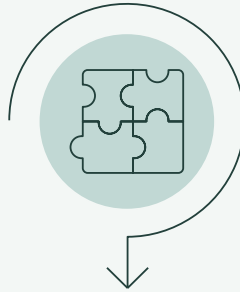
What are IEEE Standards?

The rules that allow different devices to talk to each other



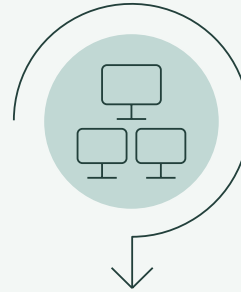
IEEE

**The Institute of
Electrical and
Electronics
Engineers.**



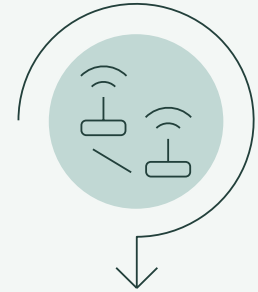
Standardization

**Ensures hardware
from different
vendors is
compatible.**



802 Committee

**The group
responsible for
Local Area Network
(LAN) standards.**

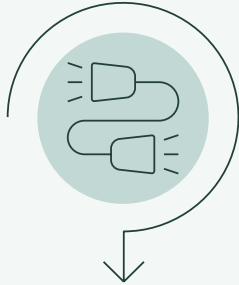


Interoperability

**Why your laptop can
connect to any
router or switch.**

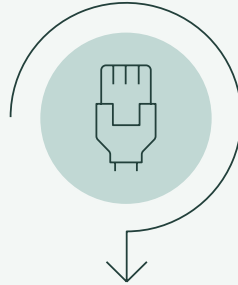
Ethernet (802.3) [Wired LAN]

The backbone of wired local area networks: A standard for physical and data link layer connectivity.



Physical Connection

Ethernet provides a reliable and high-speed physical connection between devices.



Standardized Cables

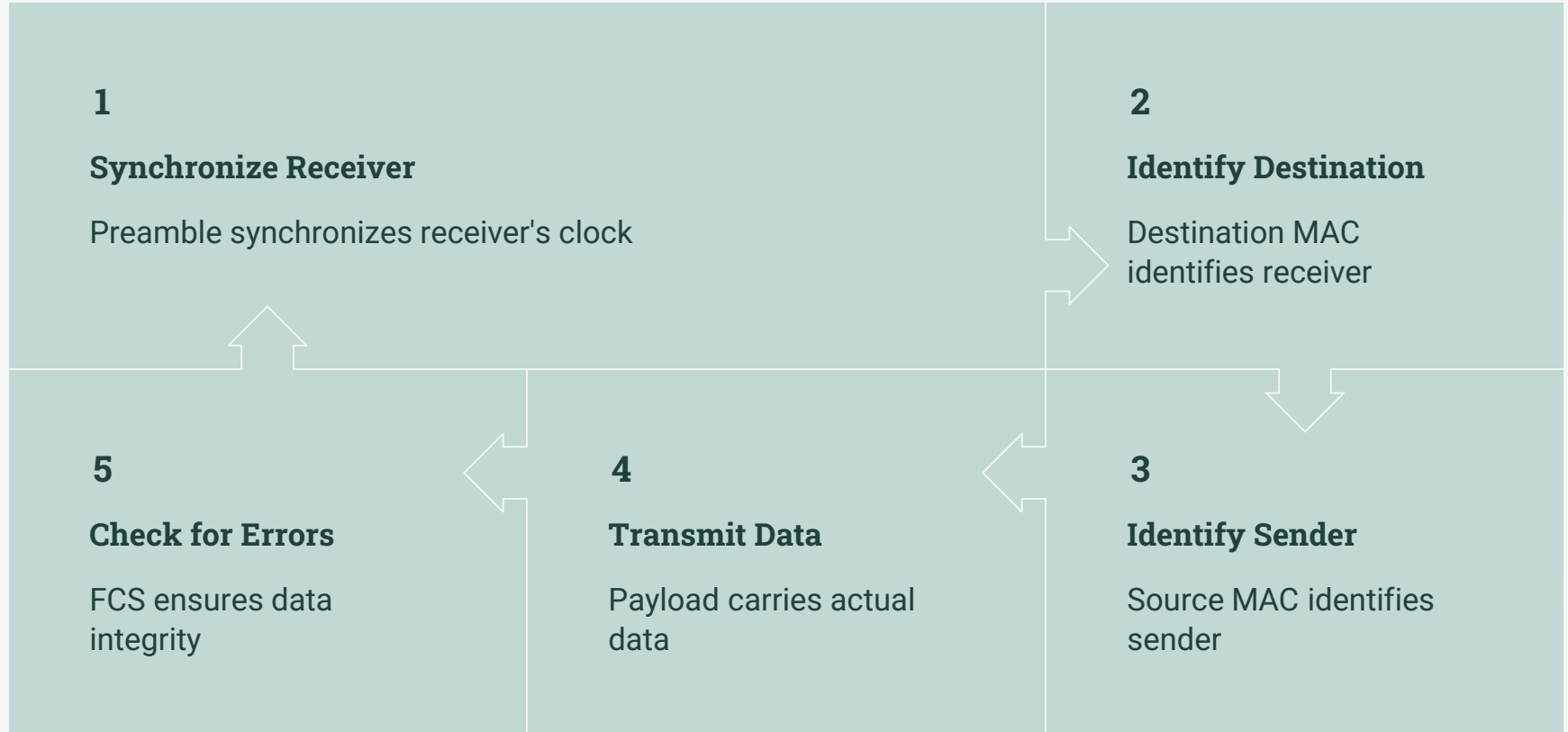
Ethernet uses standardized cables, such as Cat5e or Cat6, for data transmission.



Wide Usage

Ethernet is commonly found in homes, offices, and large data centers.

Ethernet Frame Transmission Cycle



Collision Control

How CSMA/CD keeps traffic moving on the wire

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) acts as the traffic cop for older Ethernet networks, managing how devices share the network medium.

Characteristic

Description



Carrier Sense

Listen before transmitting



Multiple Access

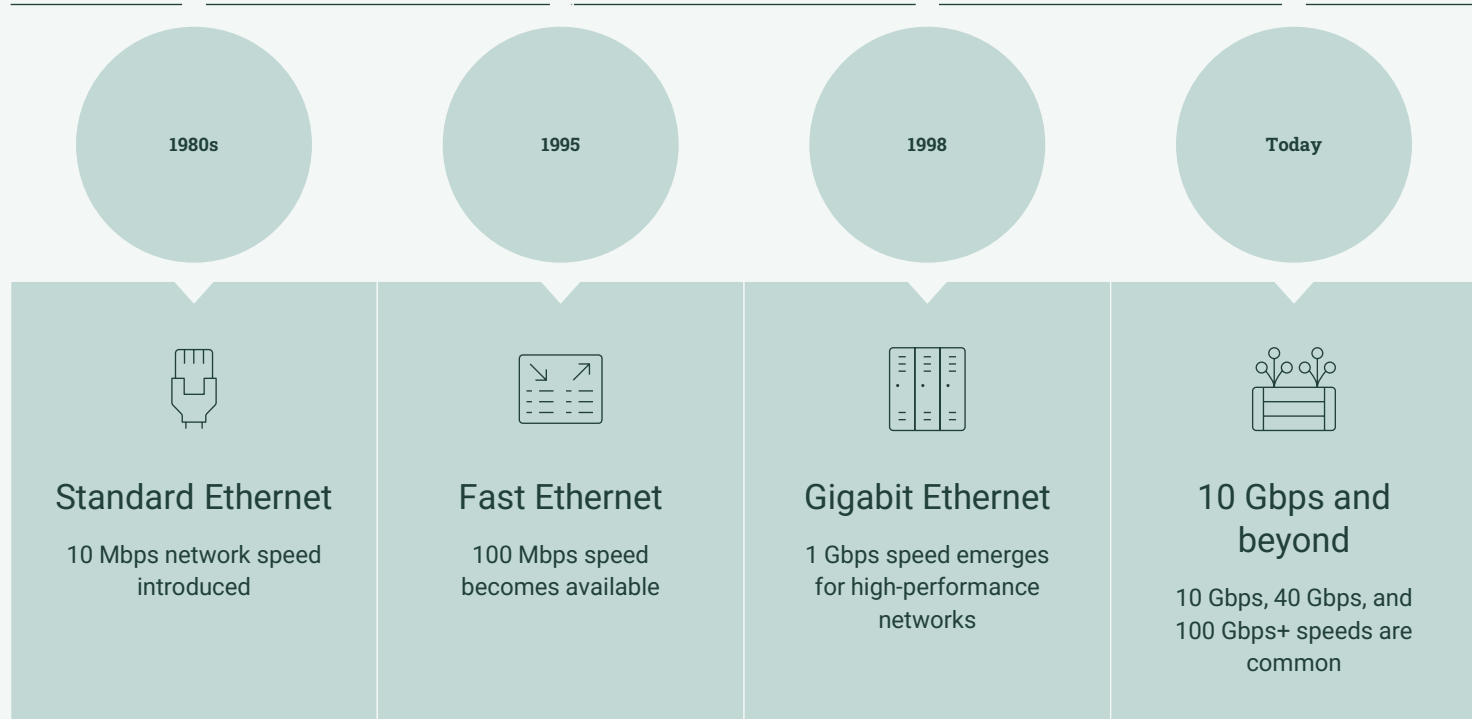
Shared wire for all



Collision Detection

Stop and retry if conflict

Evolution of Ethernet Speed



ETHERNET

Ethernet network can be classified into Four types:

- i) Standard Ethernet
- ii) Fast Ethernet
- iii) Gigabit Ethernet
- iv) Ten-Gigabit Ethernet

i) Standard Ethernet:

- This type of Ethernet can transfer data at a rate of **10 Mbps**.

ii) Fast Ethernet:

- This type of Ethernet can transfer data at a rate of **100 Mbps**.
- Fast Ethernet makes use of **twisted pair cable or fiber optic cable** for communication.

ETHERNET

iii) Gigabyte Ethernet:

- This type of Ethernet network can transfer data at a rate of **1000 Mbps**.
- Gigabit Ethernet also makes use of **twisted pair cable or fiber optic cable**.
- **48 bits used for addressing in Gigabit Ethernet**.
- The **latest Gigabit Ethernet** is a **10 Gigabit Ethernet**, which can transfer data at a rate of **10 Gbps**.
- Gigabit Ethernet was developed so that it can meet the needs of the user like faster communication network, faster transfer of data etc.

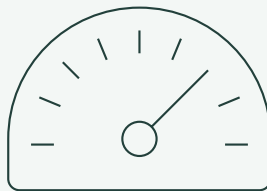
iv) Ten-Gigabit Ethernet:

- 10 Gigabit Ethernet is the recent generation and delivers a data rate of **10 Gbit/s (10,000 Mbit/s)**.
- It is generally used for **backbone in high-end applications** requiring high data rates.

Fast Ethernet (802.3u)

Breaking the 100 Mbps barrier

- 10 times faster than original Ethernet
- Backward compatible with 10 Mbps devices
- Auto-negotiation for optimal shared speed



Fast Ethernet

100 Mbps

10 times faster than original



Original Ethernet

10 Mbps

Original speed

Auto-negotiation

- Auto-negotiation is a feature in Fast Ethernet (and newer Ethernet standards) where two connected devices automatically decide the best communication settings before exchanging data.
- Example:
 - Suppose you connect: A laptop whose NIC supports 10/100/1000 Mbps; A switch that supports 10/100 Mbps
 - When the cable is plugged in, both communicate briefly before sending user data.
 - They discover:
 - Both support 100 Mbps
 - Both support Full-duplex
 - So they automatically choose:
 - ☒ 100 Mbps Full-Duplex
 - No manual configuration is needed.



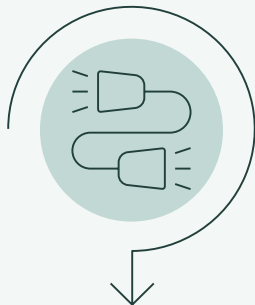
Laptop NIC card supports:
10 Mbps
100 Mbps
1000 Mbps



Switch supports:
10 Mbps
100 Mbps

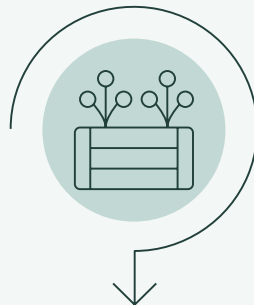
Gigabit Ethernet (802.3ab)

Powering today's high-speed LANs: 1000 Mbps standard for modern wired connections



High Bandwidth

Supports high-bandwidth tasks like 4K video and large file transfers.



Four Wire Pairs

Uses all four pairs of wires in a standard Cat5e/Cat6 cable.

Which cabling medium best suits your network requirements?

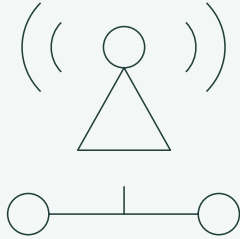
Copper (UTP)		Fiber Optic		
Inexpensive to purchase and deploy Budget-friendly infrastructure		Cost Efficiency		Higher initial investment required Premium hardware pricing
Simple and straightforward to install Easy deployment process		Installation Complexity		Requires specialized handling and expertise Complex integration procedures
Limited to 100 meters Short-range connectivity		Transmission Distance		Supports massive distances Long-range data transmission
Susceptible to electromagnetic interference Prone to signal degradation		Signal Integrity		Immune to electromagnetic interference Reliable data transmission

Ethernet Cables

Various Types of Ethernet Cables:

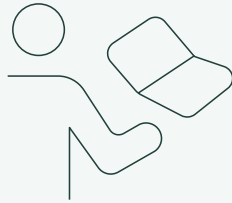
- **10Base2:** This is a **thin twisted pair** coaxial cable.
- **10Base5:** This is **thick twisted pair** coaxial cables.
- **10Base T:** This is a **twisted pair** cable which offers a speed of around **10 Mbps**.
- **100BaseTX:** This is a **twisted pair** cable and offer a speed of **100 Mbps**.
- **100Base FX:** **Fiber optic** protocol which offers a speed of **100 Mbps**.
- **1000Base SX:** **Fiber optic** protocol which utilizes a **wavelength of 850nm for multimode networks**.
- **1000Base LX:** Fiber optic protocol which utilizes a **wavelength of 1310 nm, for multimode networks and up to 1550nm for single mode networks**.

IEEE 802.11 Standard [Wireless LAN]



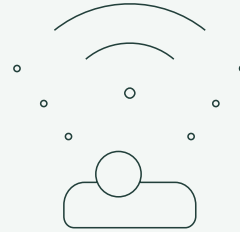
Communication Method

Devices communicate using radio waves instead of physical cables.



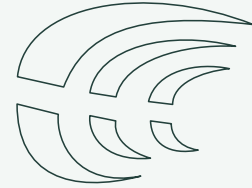
User Benefits

Mobility and convenience for users.



Medium

Shared medium (the air) instead of dedicated wires.



Brand Name

Commonly known by the brand name 'Wi-Fi'.

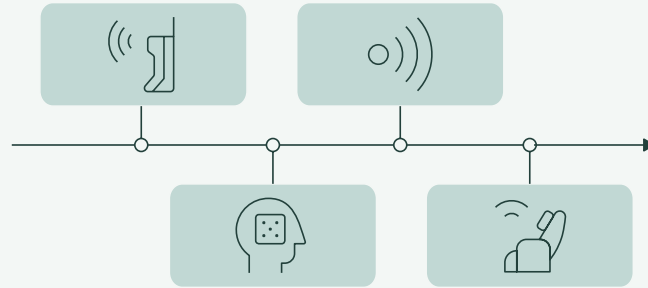
Wi-Fi Frequency Bands

Characteristic	Range	Wall Penetration	Speed	Interference
2.4 GHz	Longer	Better	Slower	More crowded
5 GHz	Shorter	Worse	Faster	Less interference
6 GHz	Ultra-high speed	Not specified	Ultra-high speed	Not specified

CSMA/CA Wireless Data Transmission Steps

Check Air
Device listens to the wireless medium to ensure it's clear

Send Data
Device transmits data to the receiver



Random Delay
If the air is busy, the device waits a random amount of time

Confirm Receipt
Receiver sends an ACK to confirm data arrived

Wireless Network Components



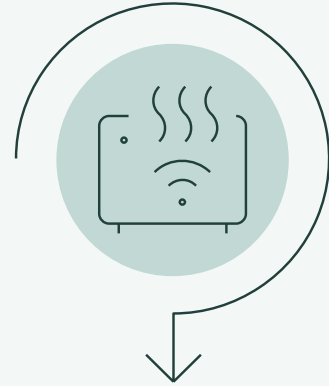
Access Point

**The bridge between
the wired and
wireless world.**



Station

**The client device
(laptop, phone,
tablet).**



SSID

**The name of the
wireless network
you see on your
device.**

Which network connection standard best suits your infrastructure needs?

Wired (802.3)

Delivers maximum speed, security, and stability

Ideal for high-demand gaming and servers



**Performance
Priority**

Wireless (802.11)



Prioritizes mobility and user convenience

Best suited for portable devices like phones and laptops

Advancing Connectivity



Wi-Fi 7

Offers extreme high throughput and lower latency for wireless networks.



Terabit Ethernet

Pushes wired speeds to 1.6 Tbps and beyond for high-performance networks.



IoT Integration

Standards evolving to support billions of small devices for interconnected systems.

A college computer lab uses a shared Ethernet cable where only one computer can transmit at a time. Which technique is most likely being used?

A. Token Passing

B. CSMA/CD

C. CSMA/CA

D. Polling

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A company upgrades from 10 Mbps Ethernet to 100 Mbps Fast Ethernet while keeping the same network architecture. What is the main benefit?

A. Longer cable

B. Higher data transmission speed

C. Wireless communication

D. Lower frequency

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A Fast
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What does it
negotiate?

A. IP Address

B. MAC Address

C. Speed and Duplex Mode

D. Frame Size

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A laptop supports 1 Gbps. The switch supports only 100 Mbps. After auto-negotiation, communication occurs at:



A. 1 Gbps



B. 100 Mbps



C. 10 Mbps



D. No communication

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Which
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Gigabit
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A. It always requires fiber.

B. It can operate over copper or fiber.

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Why does Wi-Fi use CSMA/CA instead of CSMA/CD?

- A. Wireless devices cannot reliably detect collisions while transmitting.
- B. Wi-Fi has no collisions.
- C. Wi-Fi uses optical fiber.
- D. Wi-Fi uses token passing.



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Which
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Ethernet and
Wireless
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A. Both use CSMA/CD.

B. Ethernet uses CSMA/CD (traditionally), while Wireless LAN uses CSMA/CA.

C. Wireless LAN always provides higher speed than Gigabit Ethernet.

D. Ethernet uses radio waves.

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Thank You !